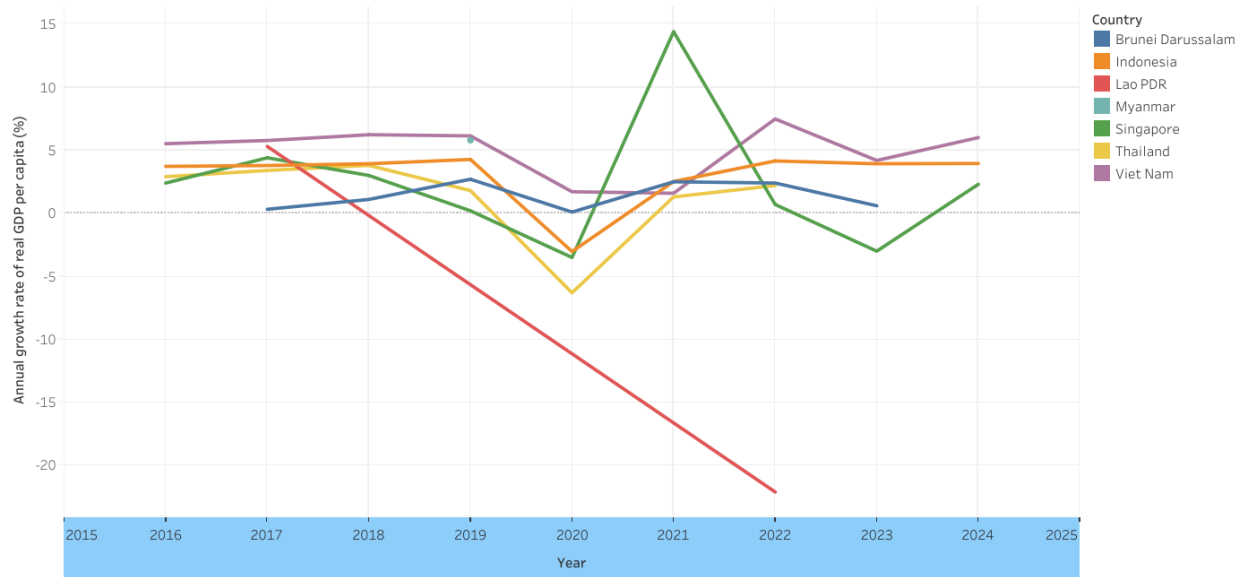


Sheet 1



Results and Interpretation

The final visualization as the figure shown above (Figure 1), provides a clear look at the economic volatility across the ASEAN region from 2016 to 2024. By transitioning the data I cleaned in Python/Jupyter Notebook over to the cloud, the trends became much easier to spot than they were in a raw spreadsheet.

Observations

- **The 2020 Economic Shock:** This is the most important trend, the synchronized V-shaped dip in 2020. This visualizes the immediate impact of the global pandemic, where tourism and trade-dependent economies like **Thailand** and **Singapore** saw their GDP growth decline into negative territory.
- **Divergent Recovery Paths:** Post-2020, the chart reveals a K-shaped recovery. While **Singapore** experienced a sharp rebound in 2021, other nations like Lao PDR show a more stagnant or downward path. This highlights the varying levels of economic resilience across the region.
- **The Role of Informal Labor:** Comparing this cloud based visualization to the raw data processed in Python/Jupyter Notebook, it is clear that during these periods of GDP contraction, the informal employment sector (which remained above 90% in some regions) acted as an important survival mechanism for the workforce when the formal economy stopped making a progress.
- **Data Gaps:** It is worth noting that certain countries, such as the Philippines, are not represented in this specific visualization. This is due to the fact that data inconsistencies or missing values in the original SDG 8 indicator source files. During the cleaning phase, these rows were excluded to maintain the integrity of the time-series analysis, to highlight the real-world challenge of standardized reporting across ASEAN.

Conclusion

Overall, this hands-on-activity demonstrated that while Python is essential for cleaning dirty or inconsistent data, cloud platforms like Tableau or SAP are what make that data understandable for decision makers. To incorporate these tools is key to tracking regional progress towards the SDB 8 goals effectively.